Cross-Age Face Verification from Mined Same-Person Social-Media Pairs

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Abstract

Cross-age face verification—deciding whether two photographs taken years apart depict the same person—is hard, and longitudinal training data are scarce. We ask a *data-centric* question: can supervision for age invariance be obtained without a manually labeled longitudinal dataset? We show that multi-photo "then/now" social-media posts supply weak *same-person* longitudinal pairs with no manual identity or age labels (a large language model parses captions and embedding clustering forms persons; we audit these components, including a small manual validation), and that fine-tuning a deliberately weak recognizer on them induces *transferable, targeted* large-gap robustness. On the external benchmark FG-NET, large-gap ($\geq$25-year) ROC-AUC rises **from 0.736 to 0.848** ($+0.112 \pm 0.001$ over three seeds; DeLong $p < 10^{-19}$), at the cost of -0.019 LFW accuracy and *substantial* low-FAR degradation—a targeted retrieval gain, not a general-verifier upgrade. Our central, bounded claim is that, within this weak/medium-backbone attribution protocol, the *data source*—not the choice among the objectives we test—accounts for most of the gain: a pair-contrastive objective matches an ArcFace-margin one (0.848 vs. 0.851). The gain is driven by the same-person pairing (caption age anchors add only $+0.009$), is mechanistically tied to removing an age shortcut, beats a real re-aging model under a gap-parity control, and transfers to an independent non-VK platform.

1 Introduction

Cross-age face verification arises in archival retrieval, account recovery, and authorized search for missing persons. The challenge lies not in discriminating among arbitrary individuals but in distinguishing *the same person across age* from *different people of similar age and appearance*. Annotating longitudinal identity pairs (one individual across many years) is costly, and such data remain scarce, which constrains supervised approaches; recent work continues to report weak cross-dataset generalization and a shortage of high-quality longitudinal data.

Hypothesis. A multi-photo "then/now" post showing one person at different ages yields $\binom{n}{2}$ positive identity pairs without manual annotation; a caption stating ages adds a secondary age anchor. The same-person pairing is the operative signal (an ablation isolates the age anchor at a marginal $+0.009$); this is a scalable, weakly/naturally-supervised signal of cross-age identity, and fine-tuning a weak recognizer on it should transfer to external benchmarks.

Contributions (bounded). (1) A weakly/naturally-supervised (no manual id/age labels) longitudinal signal mined from social posts—same-person identity pairs, with caption age anchors only a secondary cue. (2) A rigorous *attribution protocol* (one weak trainable backbone, frozen vs. fine-tuned, external evaluation) that isolates the contribution of *data* from network capacity. (3) Mutually reinforcing controls (loss family, architecture, source community, training objective, real-vs-synthetic, shortcut diagnosis, cross-platform transfer) supporting the bounded claim that, within this protocol and across the objectives tested, the data contributes more to the large-gap gain than the choice among loss families. (4) A curation pipeline with an audit of its automatic components, and the finding that naive positive counts are inflated about 2× by near-duplicate frames. The novelty is data-centric rather than architectural—a new *origin of supervision*—and should be evaluated accordingly. A supplementary technical appendix reports the controls and audits deferred for space.

2 Related Work and Positioning

Three lines of prior work are most relevant. *Discriminative age-invariant FR on labeled data:* OE-CNN (orthogonal embedding decomposition) (Wang et al. 2018), DAL (decorrelated adversarial learning) (Wang et al. 2019), and MTLFace (multi-task recognition, age synthesis and attention decomposition) (Huang, Zhang, and Shan 2021) achieve strong accuracy but are trained on datasets with *manual identity and age labels* (CACD (Chen, Chen, and Hsu 2014), MORPH (Ricanek and Tesafaye 2006), AgeDB (Moschoglou et al. 2017), FG-NET (Panis et al. 2016)). *Cross-age contrastive with synthetic samples:* CACon (Wang, Sanchez, and Li 2024) adds a synthesized cross-age sample and a triplet loss; OrdCon (Wang, Sanchez, and Li 2025) uses order-enhanced contrastive learning. *Synthetic aging for robustness:* systematic studies show synthetic aging helps only partially (Yao et al. 2024). Generic self-supervised FR (SimCLR-style augmentation (Chen et al. 2020)) lacks identity anchors across time. We instead mine the structure "one post = one person at different ages" as a label-free positive-pair generator (a per-method positioning table is in the supplement). We report on the canonical cross-age benchmarks of these methods (AgeDB-30, CALFW (Zheng, Deng, and Hu 2017), FG-NET; CALFW is a harder, age-gapped LFW (Huang et al. 2007)). The goal is *not* to surpass the absolute state of the art (we use a weak backbone for clean attribution) but to quantify the magnitude and transferability of the gain from a naturally-supervised source; Sec. 5.1 trains the canonical SOTA objective on our data directly.

3 Data: Naturally-Anchored Longitudinal Pairs

Source. Two public "then/now" multi-photo communities on VK (one platform, Russian-language context). Faces are detected and aligned with a 5-point ArcFace template (InsightFace RetinaFace (Deng et al. 2020b), `buffalo_l`). A *usable* face passes detection-score, minimum-size, blur, and single-target checks (Sec. 4). The raw mined corpus is summarized in Table 1.

Age extraction. Captions are parsed by regular expressions (Russian patterns plus the slash format "6/18/21") and by a large language model (LLM). Across all 33,697 multi-photo captions the LLM broadens and regularizes extraction relative to regex—it avoids reading calendar years or gaps as ages and recovers second ages—disagreeing on 27.2% of posts, overwhelmingly LLM-favoring; on a 100-caption manual gold subset the LLM matched the human reading on 89% vs. regex 71% (supplement). We do not treat LLM output as ground truth: downstream, explicit age anchors contribute only marginally (Sec. 5.1).

Persons and leakage-safe split. A naive "one post = one identity" undercounts: a person recurs across posts. We cluster post-groups into persons by embedding similarity (cosine $\geq$ 0.85), giving 21,848 persons from 27,487 post-groups ($\approx$20% recurrence), and split *by person* (not by post). Manual inspection showed cosine does not separate look-alikes from same-person in the 0.55–0.8 band, so the merge threshold is deliberately conservative.

Auditing the supervision. The automatic components are potential hidden label noise, audited three ways: LLM-vs-regex age agreement; within-person apparent-gender consistency as an inexpensive multi-person detector; and a *small* manual validation (one annotator) that *sanity-checks* the pipeline (age, group integrity, pair and merge precision 0.93–0.96; supplement). With a single annotator and no inter-annotator κ this *supports* but does not *certify* the supervision; a large multi-annotator audit is beyond our resources, so residual noise is acknowledged as a limitation (Sec. 7).

Curation and its largest finding. We apply per-person apparent gender/age aggregation, LLM group-integrity validation (92.8% single-person), and near-duplicate dedup within each person (cosine $\geq$ 0.97). Because the number of positives is quadratic in faces per group, removing near-duplicates cut the positive count from 65,897 to 31,865 (-52%): **about half of the raw "positives" were duplicate frames** (a single shoot, burst, or repost) inflating internal metrics. After curation the frozen internal baseline drops (our.25+ $0.705 \rightarrow 0.640$), exposing a harder, more representative benchmark; external benchmarks are unaffected. The reduction is stable across dedup thresholds 0.93–0.97 (supplement). Provenance, rights and governance fields are stored per item; a full datasheet (Gebru et al. 2021) accompanies the controlled-access release (see the Ethical Statement).

4 Method, Protocol and Statistics

Attribution protocol. Comparing an adapter on top of a *frozen* strong ArcFace against frozen ArcFace is invalid (the adapter merely deforms near-perfect embeddings). Instead we use *one weak trainable backbone* (FaceNet InceptionResnetV1 (Schroff, Kalenichenko, and Philbin 2015), CASIA-WebFace) and compare (a) frozen $\rightarrow$ benchmark metric vs. (b) soft fine-tuning of the head on our pairs $\rightarrow$ benchmark metric, with a margin contrastive loss on aligned crops. *Evaluation is on external benchmarks* (LFW, AgeDB-30, CALFW, FG-NET) for a clean attribution of transfer; the internal test (`our.*`) is a secondary in-domain diagnostic. Backbone strength is varied (Sec. 5.3) to map where the data helps. The design trades absolute performance for *causal attribution*.

Backbones and metrics. FaceNet (weak); ArcFace iResNet-50 (Deng et al. 2019) (weak, different architecture); AdaFace IR-50/IR-101 (Kim, Jain, and Liu 2022) and ArcFace iResNet-100 (strong; a clean depth control). We report LFW (10-fold accuracy), AgeDB-30 and CALFW (ROC-AUC and accuracy), and FG-NET (ROC-AUC overall and on the large-gap $\geq$25-year subset), distinguishing *threshold-free* metrics (ROC-AUC) from *threshold-based* ones (accuracy, TAR@FAR).

Statistics. The *primary endpoint* is FG-NET large-gap ROC-AUC, with the null hypothesis that fine-tuning does not improve it over frozen (gain $\leq$ 0); AgeDB-30, CALFW and the internal cross-age test are secondary, and easy LFW accuracy is a forgetting control. The headline gain is mean $\pm$ std over three seeds. We report bootstrap CIs, EER and TAR@FAR (the internal test additionally uses an identity-level, person-resampling bootstrap), DeLong's test for two *correlated* ROC curves (DeLong, DeLong, and Clarke-Pearson 1988), a paired permutation test, and a Holm correction across the stratified fairness tests (full operating-point tables in the supplement). All headline experiments use the curated data (21,427 groups, 40,586 faces, 31,865 positives; by-person split with balanced negatives).

5 Results

5.1 Main Effect and Supervision-Source Ablation

The weak FaceNet, softly fine-tuned on our pairs, raises FG-NET large-gap ROC-AUC **0.736** $\rightarrow$ **0.848** ($+0.112 \pm 0.001$), while easy LFW falls only -0.019 and AgeDB-30/CALFW move negligibly (Table 2). An ablation isolates the source: same-post pairs use no manual labels and deliver almost all of the gain; adding explicit age anchors contributes only $+$**0.009** on the 25+ bucket. The method is therefore *weakly supervised by naturally occurring same-post co-occurrence*, with age supervision marginal.

Stage	Posts	Photos	Usable faces	Persons (groups)	Positive pairs
Raw mined (2 communities)	44,609	84,147	49,606	21,848	65,897
After curation	—	—	**40,586**	**21,427**	**31,865**

Table 1: Dataset scale (raw mined) and after curation. Removing near-duplicate frames roughly halves the positive count.

Metric	Frozen	+pairs (mean ± std)	Gain	(Pre-clean gain)
FG-NET large-gap	0.7361	0.8484 ± 0.0007	$+$**0.1124**	$+0.1215$
FG-NET ROC	0.8955	0.9230 ± 0.0006	$+0.0275$	$+0.0277$
our.25+ (internal)	0.6403	0.8348 ± 0.0027	$+$**0.1946**	$+0.1677$
our.overall (internal)	0.8363	0.9163 ± 0.0009	$+0.0801$	$+0.0764$
LFW accuracy	0.9688	0.9503 ± 0.0031	-0.0185	-0.0283
AgeDB-30 ROC	0.9530	0.9462 ± 0.0013	-0.0068	-0.0155
CALFW ROC	0.9482	0.9489 ± 0.0014	$+0.0007$	-0.0030

Table 2: Main result: weak FaceNet frozen vs. +pairs, three seeds, curated data (pre-clean gain shown for reference). The gain is concentrated on the FG-NET large-gap subset.

Bootstrap confidence intervals (seed 42) confirm the primary endpoint: FG-NET large-gap rises from 0.736 [0.70, 0.78] to 0.848 [0.82, 0.88] with *non-overlapping* 95% CIs; the equal-error rate nearly halves (0.335→0.219) and TAR@FAR=1% nearly doubles (0.21→0.39). Because bootstrap-CI overlap ignores the samples the two models share, we confirm the endpoint with DeLong's test for correlated ROC curves (DeLong, DeLong, and Clarke-Pearson 1988): on FG-NET large-gap $z = 9.3$, $p = 1.4 \times 10^{-20}$ (ΔAUC 95% CI $[+0.088, +0.135]$); a 10^4-fold paired permutation test agrees ($p < 10^{-4}$). The same test shows the gain is *targeted*: AgeDB-30 decreases slightly (-0.008, $p = 4 \times 10^{-6}$) and CALFW is unchanged ($p = 0.95$). Operating points (Table 3) make the trade-off concrete: large-gap AUC and TAR@1% rise sharply, while easy-benchmark low-FAR points (LFW TAR@0.1%) degrade.

Objective invariance. Training the canonical modern-FR margin objectives—ArcFace, CosFace and SphereFace identity classification (Deng et al. 2019) (the core of OE-CNN/MTLFace)—on our naturally-supervised identities, same backbone and protocol, the contrastive, ArcFace and CosFace objectives **coincide** on the primary endpoint (0.848 / 0.851 / 0.843, within ± 0.006); only SphereFace, hardest to optimize on 2–3-shot identities without λ-annealing, trails (0.801). A noise-robust sub-center ArcFace (Deng et al. 2020a) also lands at 0.851—explicit label-noise machinery adds nothing on the *curated* pairs. We draw the bounded conclusion that, within this protocol and among the objectives tested, the *data source* accounts for more of the gain than the objective choice (Table 4).

5.2 Real Pairs Outperform Synthetic Aging

Replacing real positives with synthetically aged versions—a domain proxy and a real re-aging network (FRAN (Zoss et al. 2022))—underperforms real pairs, and at FRAN's default *wide-gap* setting even *degrades* below frozen (FG-NET large-gap 0.727; Table 5). Identity-preserving aging reproduces texture artifacts rather than genuine longitudinal variation (pose, camera, era). A native-resolution control rules out a crop artifact: re-cropping a train subset at 512 px directly from the raw images and re-aging leaves the result unchanged (0.771 vs. 0.778). A gap-parity control sharpens this: FRAN ages to a wide 33–70-year gap whereas real pairs have a median gap of 9 years; matching the synthetic gap to the real distribution lifts FRAN to **0.759**—a modest positive in line with reported synthetic-aging gains (Yao et al. 2024)—yet real pairs (0.848) still lead by $+0.089$. Real mined pairs beat even a *gap-matched* re-aging model.

5.3 Headroom and Data Scaling

The gain reproduces on FaceNet, ArcFace iResNet, and AdaFace IR-50/IR-101 (three loss families, two architectures) and *monotonically decreases with frozen strength* (Fig. 1); a clean depth control (AdaFace IR-50 → IR-101) confirms it. Strong saturated backbones do not improve and forget slightly under fine-tuning; a gentler learning rate halves the forgetting but does not convert it into a gain. The contribution is therefore specific to weak and medium backbones with cross-age headroom—a limitation we state explicitly (Sec. 7). Varying the fraction of training identities (val/test fixed), the gain saturates early: **≈96% of the full FG-NET large-gap gain is reached with only 10% of identities** (≈1,500), plateauing from 25% (Fig. 2; each point mean ± std over three seeds). The signal is data-efficient; further gains require *harder* data (larger gaps, hard positives) rather than merely more volume.

5.4 Mechanism: the Age Shortcut

Under age-matched negatives (same age bucket, different people), frozen accuracy on the internal 25+ test falls from 0.640 to 0.517 (**near the chance level of 0.5**), and real pairs restore it to **0.838** (Fig. 3)—direct evidence that frozen models discriminate cross-age pairs largely by *age*, whereas our pairs induce discrimination by *identity*. A linear probe refines this representationally: apparent age stays well decodable from the identity embedding for *all* models (≈0.63–0.65 balanced accuracy $\gg$ 0.25 chance) and changes only

Benchmark	Frozen AUC [95% CI]	+pairs AUC [95% CI]	EER (fr.→+p.)	TAR@1%	TAR@0.1%
FG-NET large-gap	0.736 [0.70, 0.78]	**0.848 [0.82, 0.88]**	0.335→0.219	0.209→0.386	0.074→0.154
FG-NET overall	0.896 [0.89, 0.90]	0.922 [0.92, 0.93]	0.186→0.151	0.502→0.562	0.315→0.308
AgeDB-30	0.953 [0.95, 0.96]	0.945 [0.94, 0.95]	0.115→0.126	0.525→0.510	0.285→0.309
CALFW	0.948 [0.94, 0.95]	0.948 [0.94, 0.95]	0.116→0.117	0.580→0.624	0.213→0.320
LFW (AUC)	0.994 [0.99, 1.00]	0.987 [0.98, 0.99]	0.032→0.052	0.940→0.869	0.858→0.575

Table 3: Operating points and 95% bootstrap CIs (FaceNet frozen vs. +pairs, seed 42); the internal test additionally uses an identity-level bootstrap (text). LFW is shown as ROC-AUC for consistency.

Objective	FG-NET l.g.	our.25+	LFW	AgeDB-30
Frozen	0.736	0.640	0.969	0.953
+pairs (contrastive)	0.848	0.838	0.946	0.945
+ArcFace	**0.851**	0.868	0.952	0.947
+ArcFace, sub-center	0.851	0.869	0.951	0.947
+CosFace	0.843	0.860	0.952	0.945
+SphereFace	0.801	0.699	0.950	0.943

Table 4: Objective comparison (FaceNet, curated): contrastive, ArcFace and CosFace coincide on the FG-NET large-gap primary endpoint; a noise-robust sub-center ArcFace adds nothing; SphereFace (no λ-annealing) trails on sparse few-shot identities.

Metric	Frozen	+real	syn-FRAN	FRAN-gm
FG-NET large-gap	0.736	**0.848**	0.727	0.759
our.25+	0.640	**0.838**	0.549	0.622

Table 5: Real vs. synthetic positives (FaceNet, curated). FRAN-gm matches the synthetic age-gap to the real-pair distribution (median 9 y).

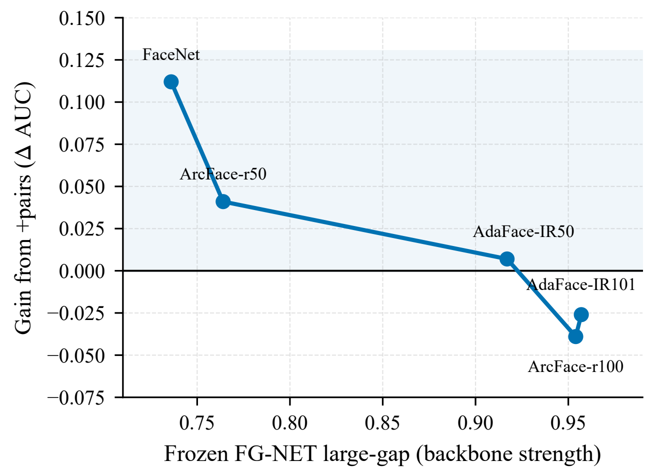

Figure 1: Headroom curve: the +pairs gain on FG-NET large-gap shrinks monotonically as the frozen backbone strengthens, crossing zero for strong saturated models.

slightly, while identity AUC rises sharply (0.836 → 0.916)—the method learns an invariant *metric* (the cosine stops relying on the age axis), not an age-free representation; explicit adversarial disentanglement reduces leakage no further (supplement).

5.5 Cross-Platform and Cross-Source Transfer

Training on one VK community and testing on the held-out other transfers the gain (external FG-NET large-gap +0.113, essentially identical to the within-source +0.112). A stronger external check is whether the *source* generalizes beyond its platform. Applying the *identical* mining protocol to an independent, non-VK community—the Reddit board r/PastAndPresentPics (same "then/now" format, same LLM parser)—and evaluating the VK-trained +pairs model with *no* Reddit data in training on 1,575 Reddit pairs raises overall ROC-AUC **0.718** → **0.749** (+0.031, near-disjoint 95% CIs). The transfer is *bidirectional*: a backbone trained on *only* a small 220-person Reddit source (a pilot, not a headline) also helps on VK (FG-NET large-gap +0.047). The gain is smaller than within-VK (a noisier, partly collage-derived set), so we read it as confirming the source generalizes beyond its platform, not as a SOTA claim. Figure 4 shows the +pairs gain stays *positive across all three* external evaluations, shrinking on harder/noisier sets.

Apparent-demographic audit. Stratifying by *apparent* gender/age of the pair anchor (estimated, not self-identified), no group degrades and every gain is significant (95% paired-bootstrap CI > 0; supplement). The largest gain is for the youngest group 0–17 (0.750 → 0.880, +0.130)—the method strengthens where the base model is weakest—corroborated on an *independent* child↔adult FG-NET subset (+0.129, non-overlapping CIs). At a matched ≈1% FMR, +pairs lowers the false-non-match rate in every apparent stratum. We avoid the word "fair" as a solved property: the source is apparent-female-skewed (76.7%) and attributes are apparent, not legal categories.

6 Discussion

The contribution is a scalable, weakly/naturally-supervised source of longitudinal supervision that teaches transferable cross-age robustness which synthetic aging cannot replace and explicit age supervision does not explain. Reinforcing controls—loss family, architecture, source community, objective, and a noise-robust margin—converge on a bounded conclusion: within this protocol, the *data source* accounts for more of the large-gap gain than the choice of objective. Crucially, the improvement is a *targeted* large-gap gain, not a drop-in upgrade to a general verifier: low-FAR operating points on easy benchmarks degrade (LFW

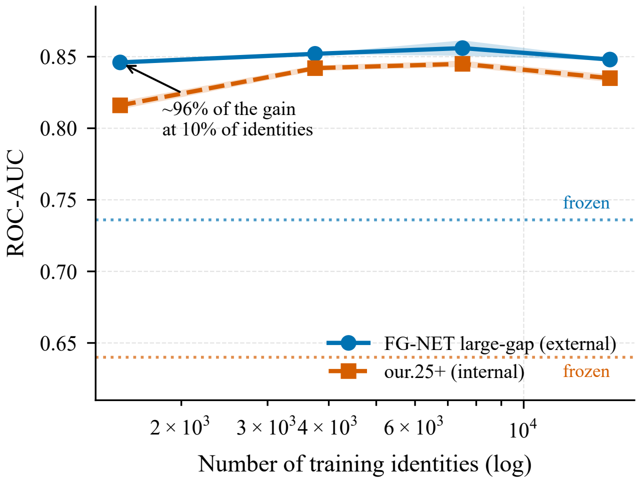

Figure 2: Data-scaling law: external (FG-NET large-gap) and internal (our.25+) ROC-AUC vs. training identities (log scale). About 96% of the gain is reached at 10% of identities; shaded bands are ±1 std over three seeds, dashed lines the frozen baselines.

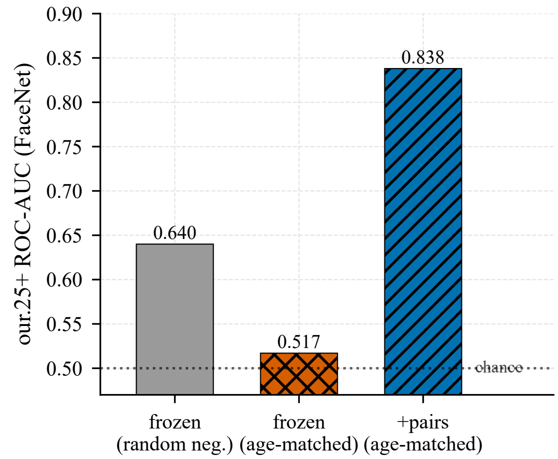

Figure 3: The age shortcut: on an identity-only test (age-matched negatives) the frozen model falls to near-chance; fine-tuning on our pairs restores discrimination by identity.

TAR@FAR=0.1% falls $0.858 \rightarrow 0.575$), so deployment should be scoped to the large-gap cross-age regime.

Scope of the claims. *Established (within this attribution protocol):* (i) the mined pairs improve large-gap cross-age verification on FG-NET (primary endpoint, non-overlapping 95% CIs) and the internal 25+ test; (ii) the gain transfers across the two source communities, to an independent child↔adult FG-NET subset, and to a non-VK platform; (iii) real pairs **outperform synthetic aging**, including under native-resolution and gap-parity controls; (iv) the gain is invariant across the contrastive, ArcFace and CosFace objectives; (v) it is mechanistically tied to **removing an age shortcut**; (vi) roughly half of the raw positive pairs were near-duplicate frames. *Not established / out of scope:* the effect is not a uniform verification gain (low-FAR operating points degrade); strong saturated backbones do not improve; a larger non-VK *training* source with multi-seed CIs remains future work; we do not reproduce a full SOTA pipeline, only its shared margin objective; the fairness audit uses apparent attributes only and lacks a multi-annotator κ.

7 Limitations

- **Backbone dependence:** the gain concentrates in weak/medium backbones; strong saturated models do not improve and slightly forget.
- **External validity:** two VK communities, one platform, one language/cultural context; apparent-female-skewed (76.7%); sparse at apparent age 45+. Transfer is shown between two VK sources, to standard external benchmarks, and to a small independent Reddit then/now test; this does not establish platform-agnostic generalization.
- **Apparent attributes only** in the fairness audit (estimated, not self-identified gender/ethnicity), and no human-audited inter-annotator κ (a large manual audit is labor-intensive and beyond our resources).
- **Weakly supervised, not label-free:** an LLM parses ages and validates group integrity and clustering forms persons—audited but a residual hidden-noise source.
- **Benchmark scale and age:** the field-standard cross-age suite (AgeDB-30, CALFW, FG-NET) fixes comparability but inherits limited scale/age; a larger non-VK longitudinal set remains useful future work.

Ethical Statement

This is a sensitive biometric study; we treat governance as a first-class component. Permitted use is restricted to controlled / consent-based / human-reviewed applications (rights-cleared archives and family photos, authorized humanitarian search, account recovery); the system outputs candidates for human review, never an automatic decision. This is a *research-only* study—nothing is deployed and *no public biometric database is released*. Forbidden: mass identification, de-anonymization, surveillance, and any processing of minors' data without a legal basis.

Ethical approval. No formal ethics-committee protocol has been filed for this study to date; given its research-only, fully de-identified design, the authors will obtain an institutional ethics review or a documented exemption if a venue or reviewer requires one (institution withheld for anonymity). *Consent.* Because the data are drawn from public posts with no feasible channel to contact subjects, individual informed consent for participation and publication was not obtained; we rely on a scientific-research legal basis with the de-identification safeguards below, and note that official API access and voluntary public posting do not by themselves authorize biometric processing (cf. GDPR Art. 9 special-category data (European Parliament and Council of the European Union 2016) and national biometric-consent law). We report this gap transparently rather than claiming full compliance.

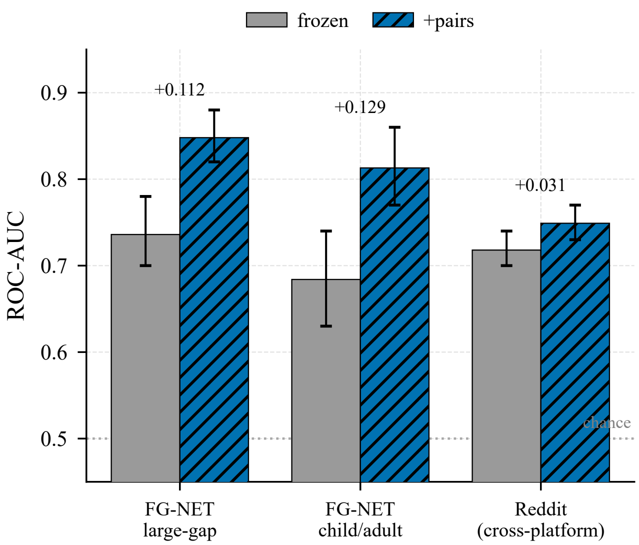

Figure 4: External generalization. The +pairs gain over frozen (95% CIs) holds across three independent evaluations—FG-NET large-gap, a child↔adult FG-NET subset, and the cross-platform Reddit set—each well above chance, largest on the in-protocol FG-NET sets and smaller (but positive) on the noisier Reddit set.

De-identification and release. We *do not release raw face images or raw posts*. The public release is restricted to code, configurations, the pseudonymized benchmark protocol and aggregate statistics. Because a face embedding is itself a comparable *biometric template* and salted identifier hashes do not render it anonymous, trained weights and derived embeddings are shared with bona-fide researchers *on request* under a brief research-use agreement (research-only; no re-identification); the de-identified benchmark dataset is likewise available on request, with formal data-protection documentation prepared if required. Platform identifiers are replaced by salted hashes; captions/comments are excluded; only an age bucket and a pseudonymous person ID are kept. Minor-age items are withheld from any release absent an explicit legal framework and verifiable guardian consent. A Data Protection Impact Assessment, a full datasheet (Gebru et al. 2021) and a takedown / deletion-on-request procedure accompany the release.

Reproducibility. We release all scripts, configurations and versioned LLM prompts; raw face images and posts are withheld, so data-mining replication is partial (we release cached LLM outputs for deterministic replay). The entire study runs on a single laptop (one 6 GB GPU). A reproducibility checklist accompanies the submission.

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